

AMAZING MART

Sales & Profitability Analysis

Power BI Data Analytics Project

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Primary Tool: Microsoft Power BI

Project Focus

Interactive analysis of sales, profit, quantity, discount, regional performance, countries, product categories, sub-categories, shipping methods, and quantity distribution.

1. Executive Summary

The AmazingMart project is a Power BI business intelligence analysis focused on understanding sales and profitability performance. The dashboard provides interactive analysis across regions, countries, product categories, sub-categories, ship modes, and time periods. The completed dashboard reports **\$2.35M in total sales, \$283.2K in total profit, 30K in total quantity**, and approximately **\$0.89K in total discount**.

2. Business Problem

AmazingMart requires a clear way to monitor sales and profitability and to understand which regions, countries, product categories, sub-categories, and shipping methods contribute most to performance. Power BI was used to transform the sales data into an interactive dashboard for comparison, filtering, trend analysis, and decision-making.

3. Business Questions

The dashboard addresses: overall sales and profit performance; sales and profit trends by region; country and product-category profitability; profit by ship mode; highest-sales sub-categories; weak or negative-profit sub-categories; and quantity distribution.

4. Data Source & Dataset Description

The project used the AmazingMart sales dataset as the source for the Power BI analysis. The data supports analysis of sales, profit, quantity, discount, geography, product categories, sub-categories, shipping methods, and dates. This report documents the Power BI project only; Excel and MySQL are not presented as separate analytical stages.

5. Data Preparation & Transformation

The dataset was prepared in Power BI before visualization. The preparation stage included checking data types, preparing date fields for time-based analysis, and ensuring numerical fields could be aggregated correctly. The prepared data was then used for measures and dashboard visuals.

6. Power BI Data Modeling

The prepared data was loaded into Power BI and organized to support analysis across region, country, product category, sub-category, ship mode, and time. The model formed the foundation for DAX measures, filters, and visual interactions.

7. DAX & Measures

DAX measures were used to produce the headline metrics and analytical values displayed in the dashboard, including Total Sales, Total Profit, Total Quantity, and Total Discount. These measures allow results to respond dynamically to report filtering.

8. Power BI Dashboard

The final dashboard contains KPI cards and analytical visuals covering sales performance by region over time, profit performance by region over time, profit by country and product category, sales and profit by sub-category, profit by ship mode, and quantity distribution. Interactive filtering supports deeper investigation.

9. Key Results & Business Insights

Metric	Dashboard Result
Total Sales	\$2.35M

Metric	Dashboard Result
Total Profit	\$283.2K
Total Quantity	30K
Total Discount	\$0.89K

Profit by Ship Mode: Economy has the highest profit contribution at approximately **\$178,658**.

Sub-Category Performance: Copiers, Bookcases, Phones, Storage, and Appliances are among the strongest sub-categories by sales.

Profit Risk: Tables produced approximately **-\$20,731** in profit and therefore requires management investigation.

Portfolio Insight: Comparing sales and profit helps distinguish products that generate revenue from those that generate healthy profitability.

10. Regional & Time Analysis

The dashboard includes sales performance by region over time and profit performance by region over time. These views allow management to identify changes in regional performance and assess whether sales growth is accompanied by profit growth.

11. Country & Product Category Analysis

The country-by-category profitability view provides a geographic and product-level perspective of profit contribution. It supports identification of markets and product categories that drive profitability and areas requiring further investigation.

12. Quantity Distribution

The quantity distribution visual examines how order quantities are distributed. This provides context about purchasing behavior and the concentration of orders at different quantity levels.

13. Business Recommendations

- 1. Investigate Tables:** Review pricing, discounts, procurement costs, and shipping economics behind the negative profit result.
- 2. Protect strong categories:** Monitor inventory and profitability for high-sales sub-categories.
- 3. Monitor regional profitability:** Evaluate profit alongside sales when assessing regional growth.
- 4. Review shipping economics:** Continue monitoring profit contribution by ship mode alongside service requirements.
- 5. Use interactive filtering:** Apply region, country, category, sub-category, and time filters before making operational decisions.

14. Dashboard Design & Reporting Approach

The dashboard places headline KPIs first, followed by comparative and diagnostic visuals. Trend, geographic, product, shipping, and distribution views provide multiple levels of business investigation within one Power BI report.

15. Project Deliverables

Power BI dashboard/report; data preparation and model; DAX measures; interactive visuals and filters; business insights and recommendations; dashboard screenshots; and this project report for portfolio/GitHub documentation.

16. GitHub Repository Structure

AmazingMart-PowerBI-Analysis/

■ ■ ■ ■ README.md

■ ■ ■ ■ Dataset/

■ ■ ■ ■ AmazingMart_Dataset.xlsx

■ ■ ■ ■ PowerBI/

■ ■ ■ ■ AmazingMart_Dashboard.pbix

■ ■ ■ ■ Dashboard.png

■ ■ ■ ■ Data_Model.png

■ ■ ■ ■ DAX_Measures.png

■ ■ ■ ■ Report/

■ ■ ■ ■ AmazingMart_PowerBI_Analysis_Report.pdf

17. Conclusion

The AmazingMart Power BI project demonstrates the transformation of transactional sales data into an interactive management reporting solution. The dashboard provides a consolidated view of sales, profit, quantity, discount, regional trends, product performance, country profitability, shipping performance, and quantity distribution. The analysis reports **\$2.35M in sales** and **\$283.2K in profit**, while highlighting strong product areas and identifying **Tables** as a negative-profit sub-category requiring further investigation.